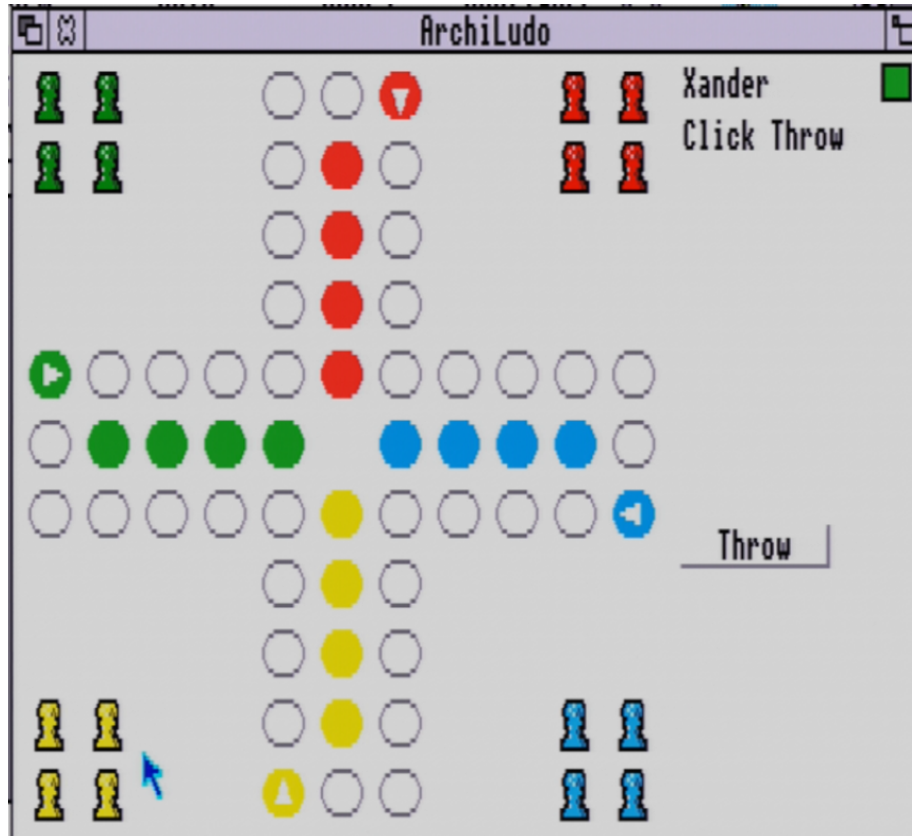


ArchiLudo



A Ludo (“Mens Erger Je Niet”) game for the Acorn Archimedes, targeting genuine 26-bit RISC OS 3.10. By Xander Mol (idreamtin8bits.com).

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Playing ArchiLudo

Four players, four pawns each, a shared ring and a short home run per player – the classic “Mens Erger Je Niet”/Ludo cross board.

Starting a game: on first launch, dismissing the startup splash screen (**OK**, or click anywhere on it) opens the **New Game** dialogue directly. Later on, click the iconbar icon to open it again. Set each of the 4 players’ name and whether they’re Human or AI-controlled; an AI-controlled player also gets a **Low/Medium/ High** difficulty picker (click to cycle) – see AI.md for what each tier actually does. Optionally click **Rules...** to pick a rule variant or adjust individual house rules (see RULES.md for the full manual), then **Start**.

Playing a turn: click **Throw** to roll. The status line tells you what to do next – click one of your own pawns on the board if more than one can legally move (if only one can, it moves automatically), or click **Throw** again on a six. Landing exactly on another pawn sends it home; reaching the exact end of your home column finishes that pawn. For an AI player’s turn, click **Continue** to step through its moves. Every time a player finishes, a win screen appears – **Continue** (offered for 1st, 2nd, and 3rd place) lets the remaining players keep going to decide the rest of the placing; the 4th and final player’s screen has no Continue, since nobody’s left to play. Every win screen also offers **New Game** and **Quit Game**.

Saving and loading: **Save Game/Load Game** from the iconbar menu open 5 fixed, renamable slots at any time during play.

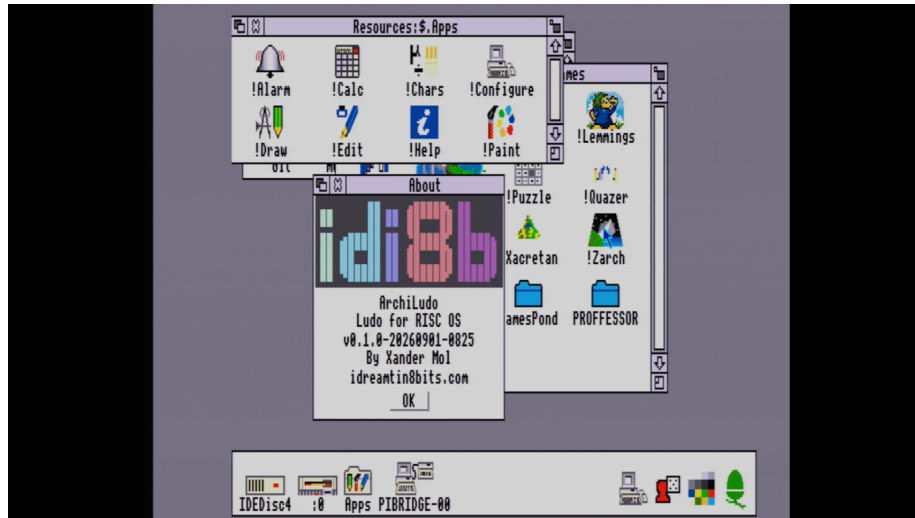
Music and sound effects: open the **Music** submenu (iconbar or window menu) to turn music **On/off**, turn **SFX** on/off independently, or pick a **Track**. See QTM.md for the full music/SFX manual (what each track and effect is, and Track 3’s one limitation).

Rules and variants: ArchiLudo ships 3 rule presets (Mens Erger Je Niet, Ludo, Pachisi-style) and 8 individually-adjustable house-rule toggles. See RULES.md for the complete rules manual.

Interface manual

All screenshots below are from a real Acorn A305 (ARM3, 4MB) over a PiEconet-Bridge, not an emulator – disc name PIBRIDGE-00 in the iconbar is the giveaway. This section walks every screen and dialogue in the order a player actually meets them; see “Playing ArchiLudo” above for the accompanying prose manual.

Splash screen





Shown once automatically on startup (and reachable again afterwards from the iconbar/window menu's **About** entry). Click **OK**, or anywhere on the window, to dismiss it – the very first time, this leads straight into the **New Game** dialogue below.

New Game dialogue

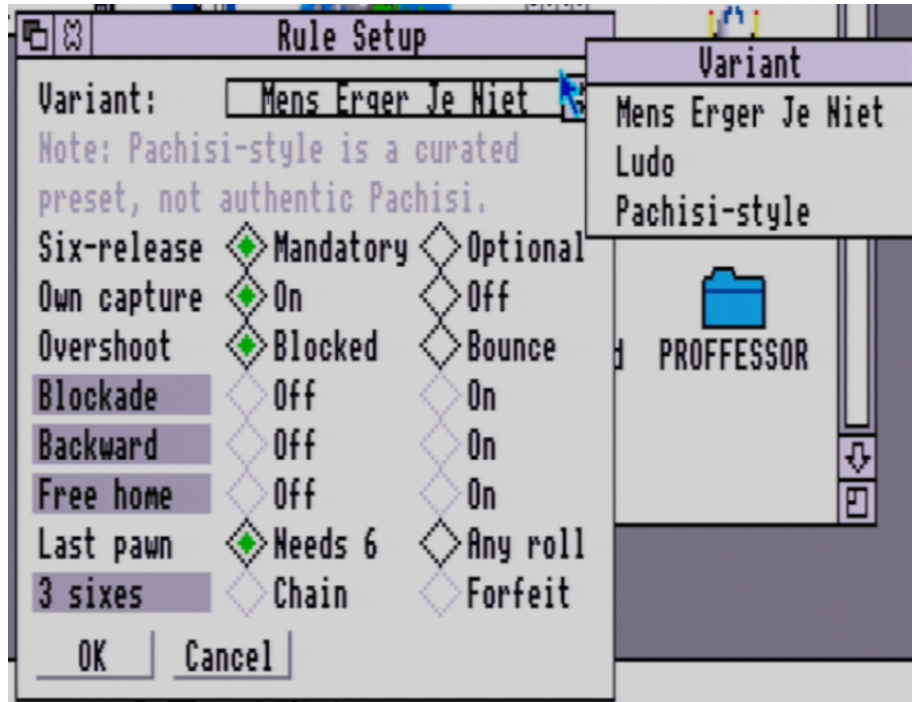


Set each of the 4 players' colour-coded name field and whether they're **Human** or **AI** (click to toggle). An AI-controlled player also gets a difficulty picker cycling **Low**/**Medium**/**High** (shaded out for a Human player, as GREEN's is here) – see AI.md for what each tier actually does.



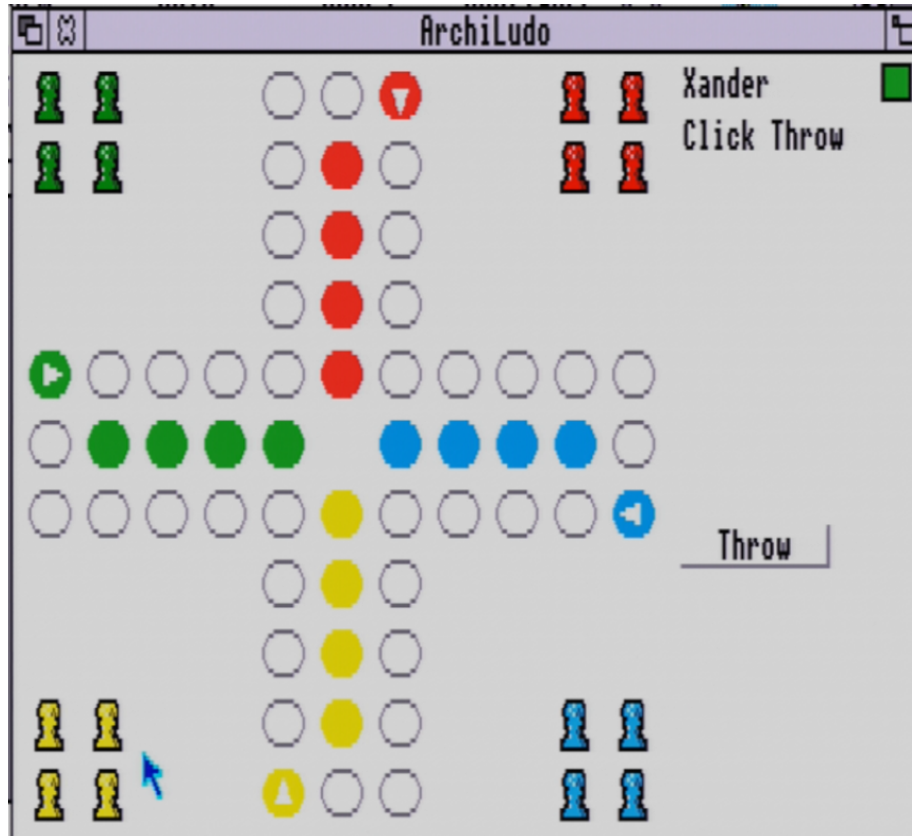
Any name field is freely editable (GREEN renamed to “Xander” here). **Rules...** opens the Rule Setup dialogue below; **Load** jumps straight to the Load Game dialogue instead of starting fresh; **Start** begins play with the configured players and whatever rules are currently selected.

Rule Setup dialogue



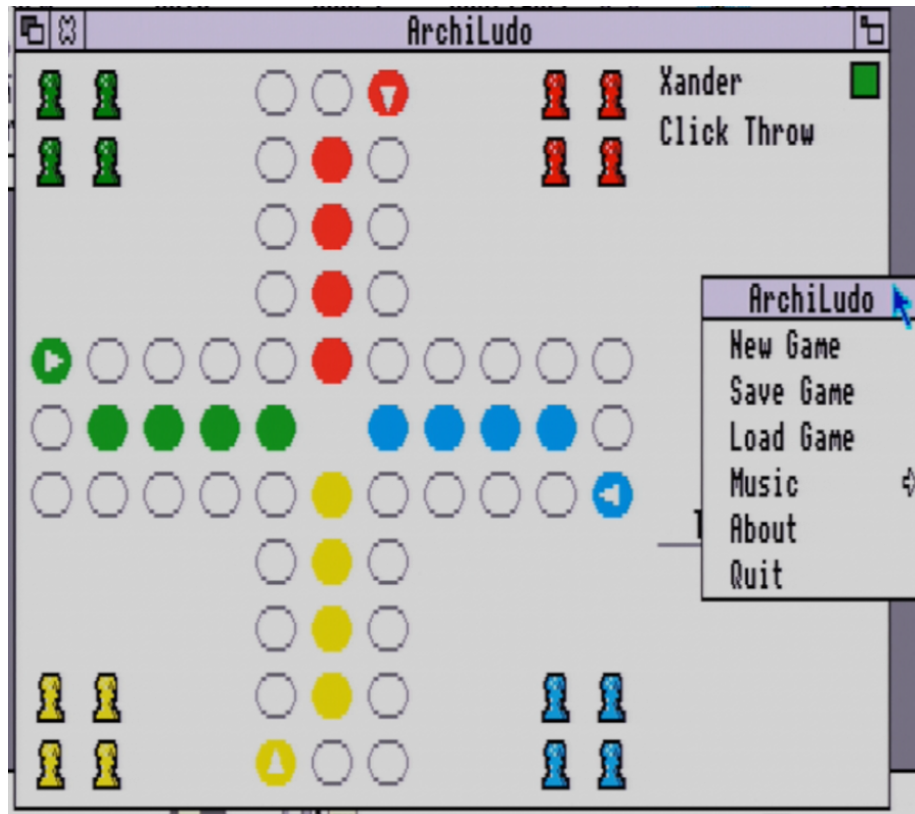
Reached via **Rules...** on the New Game dialogue. The **Variant** selector picks one of the 3 curated presets (Mens Erger Je Niet, Ludo, Pachisi-style) and fills in every toggle below to match it; any toggle can then be adjusted individually, which is what actually happens under the hood – a variant is just a starting point, not a locked mode. See RULES.md for exactly what each toggle changes about play.

The game window

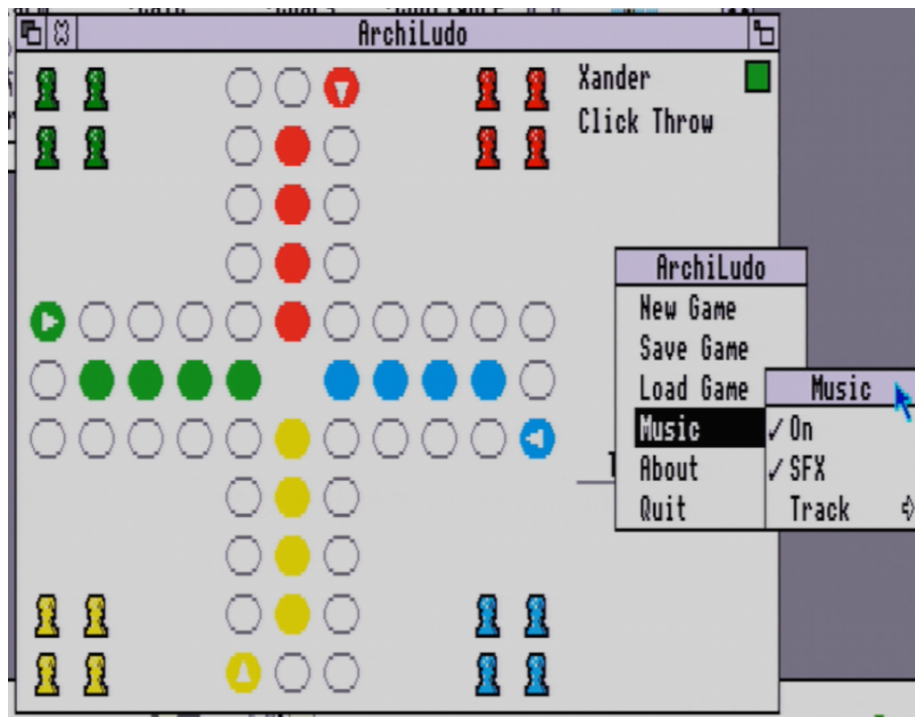


The board (also shown at the top of this README) fills most of the window: the shared 40-square ring, each player's 4-pawn home base in a corner, and their short home column running in from the ring to the centre. The panel on the right shows the current player's name and what to do next (**Click Throw**, or a prompt to pick a pawn), plus a solid colour swatch marking whose turn it is. Click **Throw** to roll; click one of your own pawns on the board to move it, if more than one legally can (a single legal pawn moves automatically without a click).

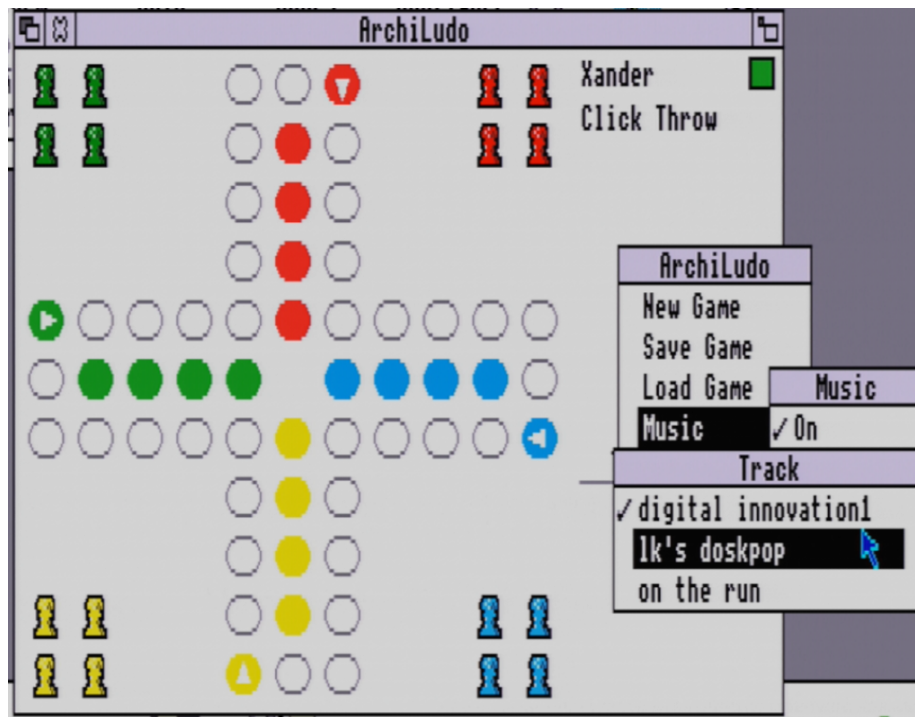
Menus



Opened from either the iconbar icon or a click on the game window itself (both show the same menu). **New Game** and **About** reopen the dialogues covered above; **Quit** closes ArchiLudo.

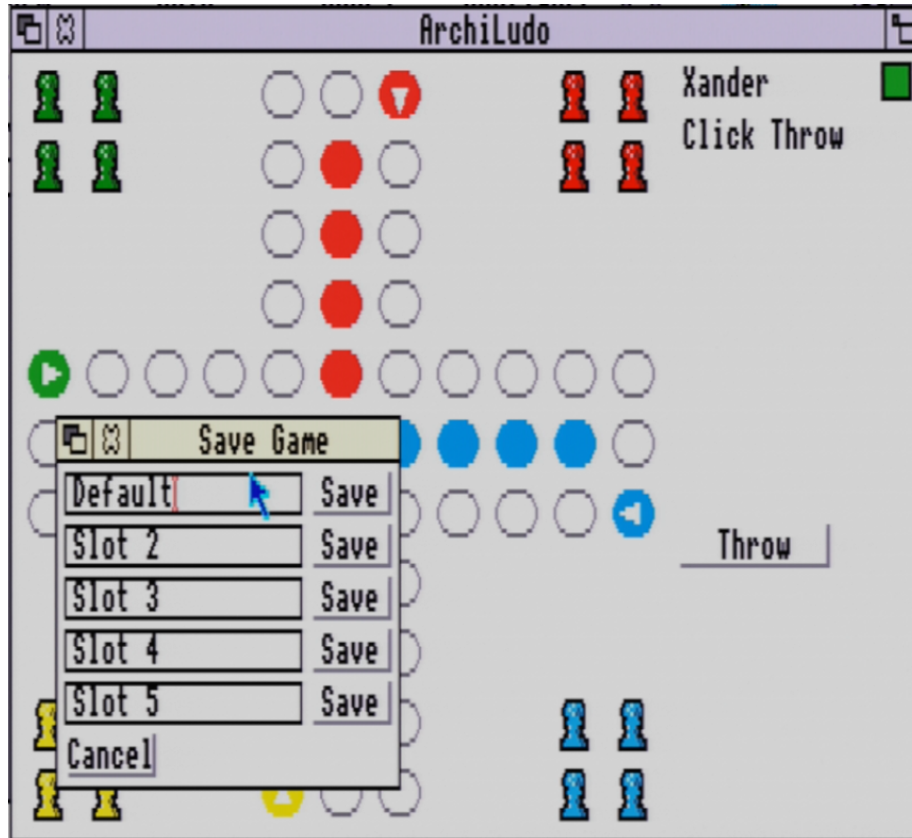


On and **SFX** are independent ticked toggles – music can play with effects muted, or vice versa. **Track** opens a further submenu (below) to pick which of the 3 bundled tracks plays.

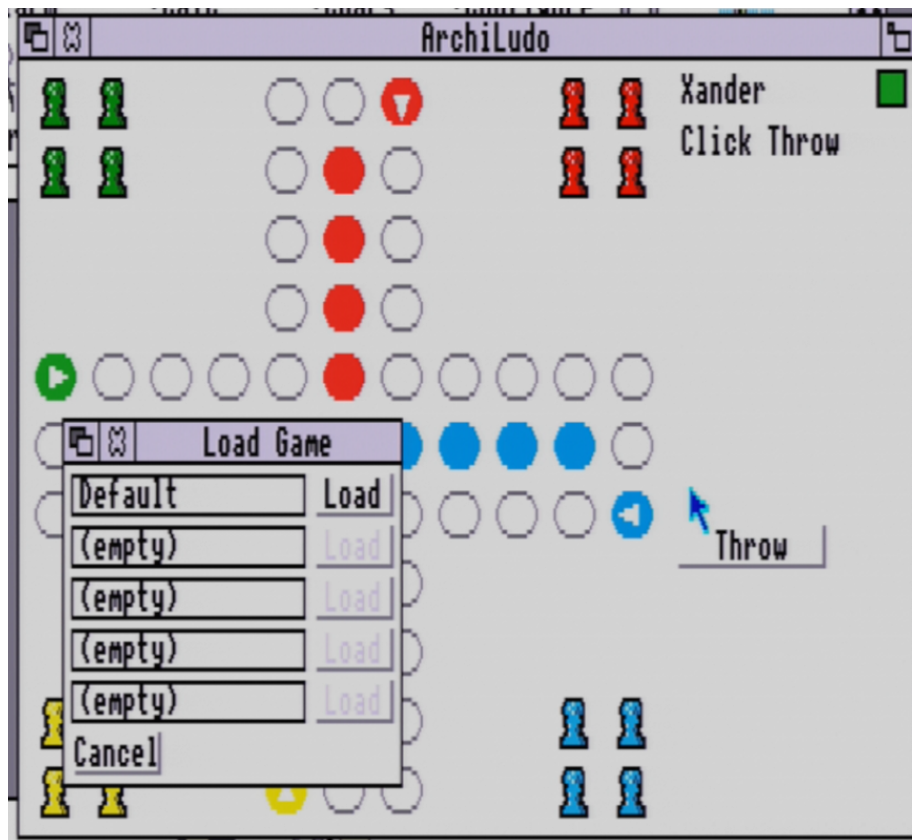


A tick marks the currently-playing track; clicking another switches to it immediately. See QTM.md for what each track and sound effect is, and Track 3's one limitation.

Saving and loading

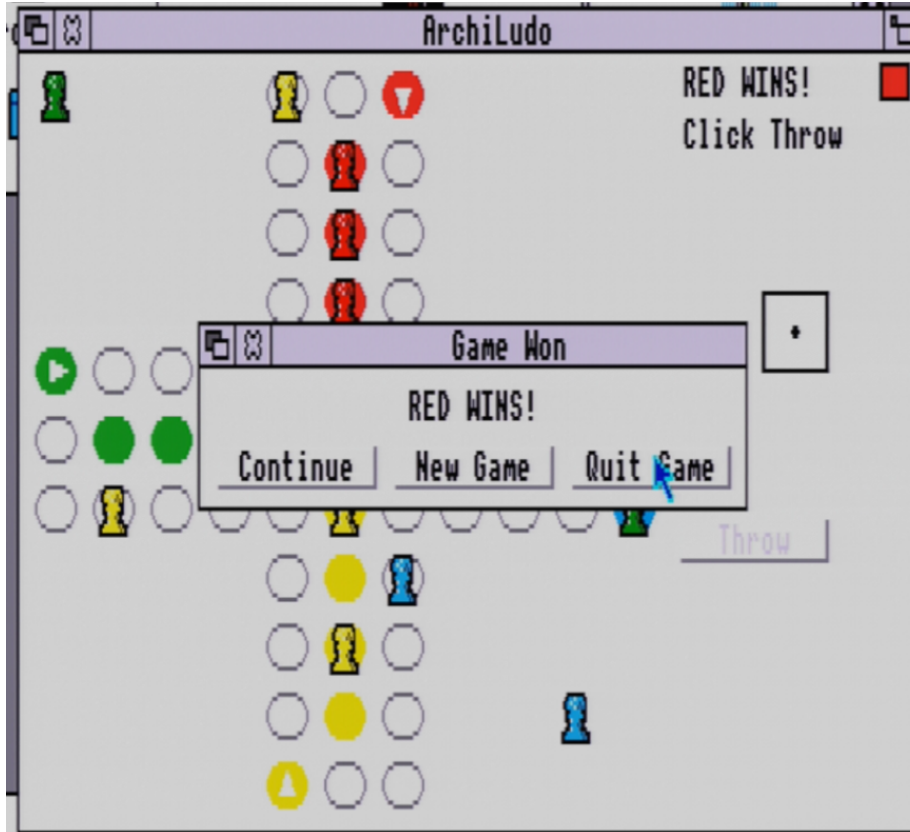


5 fixed slots, each independently renamable by editing its text field before clicking its own **Save** button – saving never prompts to overwrite, since all 5 slots are equally “yours” (no distinction between a fresh slot and one already in use).



An empty slot shows (empty) with its **Load** button shaded, exactly mirroring the Save dialogue's 5 slots. Reachable both from the iconbar/window menu and directly from the New Game dialogue's **Load** button.

Winning



A win screen (with fanfare) appears the moment any player finishes all 4 pawns, showing “<name> WINS!” for the first player home and “<name> ended 2nd/3rd/4th” for each place after that. **Continue** (offered for every place except the last) lets the remaining players keep going to decide the rest of the finishing order; every win screen also offers **New Game** and **Quit Game**.

Installation

Requires a real or emulated Acorn Archimedes running RISC OS 3.10 or later (ARM2 or ARM3). Download a release (ArchiLudo-vX.Y.Z-*.zip or .adf) and:

- **Zip**: extract with SparkFS (or any RISC-OS-aware zip tool) – filetypes are preserved. Double-click the extracted !ArchiLudo application directory to run.
- **Disc image** (.adf, ADFS “D” format, 800KB): write it to a real 3.5” floppy, or mount it directly in an emulator that supports raw ADFS images. It contains just the release zip, correctly filetype – extract it the same

way as above once it's accessible from RISC OS.

- **Real classic-Econet hardware** (a PiEconetBridge or similar old-style fileserver): rename the downloaded file to 10 characters or fewer with no dot before transferring it (e.g. **ArchiZip** for the zip, **ArchiADF** for the disc image) – such filesevers silently truncate longer names in a way that makes the file unreadable, not just renamed. Arculator's hostfs and a plain download/extract on Windows/Mac/Linux have no such limit.

No installer is needed beyond copying the **!ArchiLudo** directory wherever you want it – RISC OS applications are self-contained directories, not registry entries or system-wide installs.

Building from source

Prerequisites

Tool	Purpose	Install
ArchieSDK	ARM2-targeting cross-compiler (GCC 8.5.0), bundled OSLib	<code>git clone https://gitlab.com/_targz/archiesdk.git</code> <code>~/riscos-dev/archiesdk</code> <code>&& cd</code> <code>~/riscos-dev/archiesdk</code> <code>&& ./build.sh</code>
Arculator	Acorn Archimedes emulator, for testing	install anywhere; point <code>ARCULATOR_HOSTFS</code> in <code>.env</code> at its <code>hostfs</code> folder (e.g. <code>D:\Retro\Acorn\Arculator_V2.2_Windows\hostfs</code> on Windows)
pandoc	optional: README.md -> README.pdf	<code>sudo apt install pandoc texlive-xetex texlive-latex-extra</code>
Python 3 + Pillow	<code>tools/riscos_sprite.py</code> , the PNG->Sprite converter	<code>pip install Pillow</code>
sshpass	optional: <code>make deploy-pibridge</code> (real-hardware deploy)	<code>sudo apt install sshpass</code>

.env setup

Copy `.env.example` to `.env` and set:

```
ARCHIESDK = /home/xahmol/riscos-dev/archiesdk
ARCULATOR_HOSTFS = /mnt/d/Retro/Acorn/Arculator_V2.2_Windows/hostfs
```

PIBRIDGE_USER/PIBRIDGE_HOST/PIBRIDGE_PASS/PIBRIDGE_PATH are also needed for `make deploy-pibridge` (a real PiEconetBridge target) – see `.env.example` for the full set. `.env` is gitignored – never commit it.

`.env` is read only by `make` (a plain `-include .env` in the Makefile), not by your shell – if you use VS Code, `.vscode/c_cpp_properties.json` points its IntelliSense `compilerPath` at `${env:ARCHIESDK}`, which needs `ARCHIESDK` exported as a real environment variable separately (e.g. `export ARCHIESDK=/path/to/archiesdk` in your shell profile, or launch code `.` from a shell where you’ve already exported it).

Make targets

Target	Effect
<code>make / make all</code>	build the <code>build/!ArchiLudo</code> application directory
<code>make test</code>	build and run the game-logic unit tests with the host compiler (no ArchieSDK/Arculator needed)
<code>make clean</code>	remove <code>build/</code>
<code>make deploy</code>	copy <code>build/!ArchiLudo</code> to the Arculator hostfs folder
<code>make deploy-pibridge</code>	deploy to a real PiEconetBridge over SSH (password auth)
<code>make zip</code>	versioned release archive (<code>build/ArchiLudo-vX.Y.Z-<timestamp>.zip</code>), RISC OS filetype preserved
<code>make disk</code>	ADFS “D” format (800KB) disc image – the app and a plain-text README only (not <code>README.pdf</code> , too big to fit once it embeds every screenshot), correctly filetype
<code>make assets</code>	regenerate the pawn sprites and app icon from their Python generators
<code>make export-sprites / make import-sprites</code>	round-trip shipped sprites to hand-editable PNGs and back
<code>make docs</code>	regenerate <code>README.pdf</code> via pandoc
<code>make asm</code>	emit generated ARM assembly for the current sources into <code>build/</code>

See the “Installation” section above for the 10-character filename limit on real classic-Econet hardware – `make zip/make disk` deliberately keep the full version+timestamp in their own output filenames so multiple local builds can be told apart.

Testing: boot `configs/ArchiLudo-ARM3-4MB.cfg` (matches real ARM3/4MB hardware) or `configs/ArchiLudo-ARM2-1MB.cfg` (stock ARM2/1MB compatibility check) in Arculator with the RISC OS 3.10 ROM, then double-click ! ArchiLudo from HostFS via the Filer (see BUILDCHAIN.md’s “Application directory” section for its structure).

Changelog

Release notes, newest first. See the GitHub releases page for downloads of any version.

v1.0.1

- **Dice-roll fix:** `srand()` was never called anywhere – ArchieSDK’s `rand()` (a plain LCG, `SDK/src/libc/stdlib/rand.c`) starts from a hardcoded seed of 0 every launch, so every single game rolled the exact same dice sequence. Now seeded at startup and again when the New Game dialogue’s Start button is clicked, from `rand()`’s own current state mixed with `time(NULL)` and `os_read_monotonic_time()` – the second seeding point matters more in practice, since the time between launch and a player actually clicking Start is driven by human reaction time and differs every session, unlike the fairly fixed OS-boot-to-launch duration the startup seed alone captures.
- **Dice-parity fix:** that same LCG’s low-order bits are weak (an odd multiplier and odd increment make the low bit toggle on literally every call), so a plain `rand() % 6` made successive rolls’ parity alternate 100% of the time even though the 1-6 distribution looked uniform – verified by simulating the exact formula on the host over 600,000 rolls. Fixed by using the healthier high-order bits instead. See `GAME_LOGIC.md`’s “Dice randomness and seeding” section for the full writeup.
- **GitHub social preview image:** a new `tools/gen_social_preview.py` composites a screenshot, the idi8b logo, and version/description text into the repo’s social preview banner.

v1.0.0

First formal release – feature-complete, confirmed working both in Arculator and on a real, upgraded Acorn A305 (ARM3, 4MB) over a PiEconetBridge.

- **Documentation restructuring:** a new `TOOLS.md` manual now covers every standalone `tools//assets/*.py` script (usage, requirements, gotchas) in one place; `GRAPHICS_TOOLING.md`’s sprite-format/rendering-approach content moved into `ARCHITECTURE.md` instead of staying a separate file, and `BUILDCHAIN.md` now points to `TOOLS.md` rather than duplicating each script’s own mechanics.
- **Cleanup:** removed dead code – a vestigial, never-actually-used sample-format conversion path in `lib/qtm.c` and the 6 one-shot SFX files it was

the only reason to ship separately in the app directory (embedded into the music tracks' own MOD sample tables now, not shipped standalone); and a whole superseded pawn/dice art generator (`assets/generate_placeholder_art.py`) whose output nothing actually loaded any more, found while writing `TOOLS.md`. Debug file-logging (`debug_log()`) is now an opt-in build flag (`make DEBUG_LOG=1`) rather than always compiled in, so a default build carries none of its tracing strings or per-call file I/O.

- **Captured-pawn animation:** a displaced pawn (capture, own-pawn collision) now slides back to its home base instead of teleporting there instantly; the capture SFX was also replaced with a more expressive sound after two earlier tries read as too quiet or too flat in live testing – see `QTM.md`.
- **Full placement flow:** a win screen (with fanfare) now appears for every player who finishes, not just the first – “WINS!” for 1st, “ended 2nd/3rd/4th” for the rest, each offering Continue (except the last), New Game, and Quit Game.
- **AI difficulty tiers:** Low/Medium/High per AI-controlled player, picked in the New Game dialogue – Low takes only the objectively decisive moves (win/finish/capture/progress), Medium is the original full heuristic, High adds a genuine one-ply lookahead. See `AI.md`.
- **Launch flow:** the New Game dialogue now opens automatically right after dismissing the startup splash screen.
- **Distribution pipeline:** a filetype-preserving release zip (`make zip`), a from-scratch ADFS disc-image builder (`make disk`), and a real-hardware deploy target over Econet (`make deploy-pibridge`) – all verified against real hardware, not just Arculator, including finding and fixing a 10-character filename limit on classic-Econet filesystems and a line-ending mismatch in the bundled plain-text `README`.
- **QTM audio:** background music (3 selectable tracks) and 6 one-shot sound effects, embedded as MOD instrument samples and triggered via `QTM_PlaySample`, independently toggleable, with loudness-normalized SFX. Confirmed working on real hardware.
- **Save/load rework:** from a PRM-correct but non-functional (on Arculator's HostFS) drag-and-drop design to 5 fixed, renamable save slots.
- **Multi-rule-set / house-rule variant system:** 3 curated presets (Mens Erger Je Niet, Ludo, Pachisi-style) plus 8 independently toggleable house rules, a Rule Options dialogue, and AI support for all of them – see `RULES.md`.
- **Win/continue flow:** a win no longer ends the game outright – players can continue for runner-up order or start a new game.
- **Sprite art pivot:** from `os_plot` primitives to real `Wimp_PlotIcon`-based pawn sprites and a proper application icon, plus a hand pixel-editing round-trip workflow for touching up art in an external editor.
- **AI opponents and player setup:** a “New Game” dialogue (names, Human/AI toggle per player) and a first AI opponent adapted from the original GeoLudo edition's own algorithm.

- **Animation and redraw overhaul:** flicker-free small-region animation (dice roll, pawn slide, pulsing highlights) via careful `Wimp_UpdateWindow` usage.
- **Application directory packaging:** a real `!ArchiLudo` app directory with a custom icon, replacing an earlier flat hostfs layout.
- **Phase 1: playable core loop:** the WIMP game window, the real Mens Erger Je Niet board (ported from the original GEOS edition’s own coordinate tables), click-to-move, dice, capture/win detection.
- **Build environment:** the ArchieSDK toolchain, Arculator test profiles, and this project’s full documentation set established.

Development history

ArchiLudo isn’t this game’s first outing – it’s the latest in a line of ports going back over three decades, all by the same author. Full credits and technical detail for each generation are in `CREDITS.md`’s “Porting source” section; this is the narrative version.

- **1992:** the original, written in Commodore BASIC 7.0 for the Commodore 128 in 80-column mode – “Mens Erger Je Niet” (Dutch for “Don’t Get Angry, Man”, the house-rule variant of Ludo this whole family of ports implements), by Xander Mol under the “XAMA Software” name. The listing itself still carries its original 1992 copyright banner. Source and a working D64 disk image are preserved at `C128/BASIC Original` in the `xahmol/ludo` repository.
- **2020:** rewritten for the Oric Atmos, in BASIC.
- **2021:** rewritten a further three times – the Oric Atmos version again, now in C via CC65; a new C port for the TI-99/4a via TMS9900-GCC; and a new C port for the Commodore 128 via CC65.
- **2023: GeoLudo,** a GEOS edition sharing one executable across GEOS on the Commodore 64, Commodore 128, and Commodore Plus/4 – `xahmol/ludo/GEOS`. This is ArchiLudo’s own direct porting source: its board geometry and ring/home-column layout, its pawn and die-face artwork, and its first AI opponent all trace back to GeoLudo specifically (see `docs/ARCHITECTURE.md`’s GeoLudo-to-ArchiLudo mapping table), even though ArchiLudo’s own rules engine (`src/game_logic.c`) is a clean reimplement rather than a line-for-line port (see `docs/GAME_LOGIC.md` for why).
- **ArchiLudo:** this project – a native WIMP port for the Acorn Archimedes, targeting genuine 26-bit RISC OS 3.10. The house rules the 1992 original encoded by hand are now the configurable `LUDO_VARIANT_MEJN` preset among three curated variants (see `RULES.md`), and its own multi-decade thread continues in this README’s Changelog above.

Documentation

See `docs/` for the full project documentation: `ARCHITECTURE.md` (layering, directory structure, porting notes), `RULES.md` (the full player-facing rules manual), `GAME_LOGIC.md` (rules engine API), `BOARD_LAYOUT.md` (board geometry), `AI.md` (AI design), `QTM.md` (music/SFX manual and audio API), `TOOLS.md` (every standalone asset/build tool, its usage, and gotchas), `BUILDCHAIN.md` (toolchain, Makefile, distribution), `OSLIB.md` and `LIBARCHIE.md` (the libraries this project builds against); plus `riscos_wimp_reference.md` for the WIMP/SWI/ message API reference, and `CREDITS.md` for everything this project is built on or derived from.

Credits and licence

See `CREDITS.md` for everything ArchiLudo is built on, ported from, or otherwise sourced from. Licensed under the GNU GPLv3.